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UniSec

Medium Voltage Switchgear
PT MULTIMAS NABATI ASAHAN
MVD-01 20kV MAIN SUBSTATION
SCHEMATIC DIAGRAM
WBC OUTGOING
H02

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			COVER SHEET MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM WBC OUTGOING		Scale		= H02
				Prepared	YR						Customer	PT Multimas Nabati Asahan	& 302
00	Issued for Approval	02.01.19	YR	Checked	AW						Drawing Nbr.	IDUNISEC-BE318042-301	Sh. 1
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042			PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 31

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F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM WBC OUTGOING	Scale		= H02	F
					Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 302	
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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS														A
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)														
	DESIGNATION				DESCRIPTION										
	-AA				MULTIFUNCTION UNIT (CENTRAL UNIT)										
	-BAD1,-BAD4				CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1										
	-BAD2,-BAD5				CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2										
	-BAD3,-BAD6				CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3										
	-BAR				VOLTAGE PROTECTION RELAY										
	-BAS1				VOLTAGE SENSOR LOCATED ON PHASE L1										
	-BAS2				VOLTAGE SENSOR LOCATED ON PHASE L2										
B	-BAS3				VOLTAGE SENSOR LOCATED ON PHASE L3										B
	-BAT1				VOLTAGE TRANSFORMER LOCATED ON PHASE L1										
	-BAT2				VOLTAGE TRANSFORMER LOCATED ON PHASE L2										
	-BAT3				VOLTAGE TRANSFORMER LOCATED ON PHASE L3										
	-BAT4				VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1										
	-BAT5				VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2										
	-BAT6				VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3										
	-BCD				DIFFERENTIAL PROTECTIVE RELAY										
	-BCF				FEEDER PROTECTION RELAY										
	-BCG				GENERATOR PROTECTION RELAY										
C	-BCM				MOTOR PROTECTION RELAY										C
	-BCN				NEUTRAL (RESIDUAL) CURRENT TRANSFORMER										
	-BCP				TRANSFORMER PROTECTION RELAY										
	-BCR				CURRENT PROTECTION RELAY										
	-BCS1				CURRENT SENSOR LOCATED ON PHASE L1										
	-BCS2				CURRENT SENSOR LOCATED ON PHASE L2										
	-BCS3				CURRENT SENSOR LOCATED ON PHASE L3										
	-BCT1				CURRENT TRANSFORMER LOCATED ON PHASE L1										
	-BCT2				CURRENT TRANSFORMER LOCATED ON PHASE L2										
	-BCT3				CURRENT TRANSFORMER LOCATED ON PHASE L3										
D	-BCT4,-BCT7				CURRENT TRANSFORMER LOCATED ON PHASE L1										D
	-BCT5,-BCT8				CURRENT TRANSFORMER LOCATED ON PHASE L2										
	-BCT6,-BCT9				CURRENT TRANSFORMER LOCATED ON PHASE L3										
	-BCZ				DISTANCE PROTECTION RELAY										
	-BEF				FREQUENCY PROTECTION RELAY										
	-BER				SUPERVISION RELAYS										
	-BES				SYNCHRONIZING RELAY										
	-BET				THERMAL PROTECTION RELAY										
	-BGA1				POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT										
	-BGA2				POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)										
E	-BGA3				POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT										E
	-BGA4				POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)										
	-BGB1...-BGB3				POSITION SWITCHES OF CIRCUIT-BREAKER										
	-BGB4				POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS										
F															F

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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS						A	
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							
	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							
	-BGK		POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							
B	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST						B	
	-BGS1...-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
C	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3						C	
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
	-BGV2		POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							
	-BM		HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							
D	-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK						D	
	-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
	-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
	-BR		FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							
	-BT		THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
	-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							
	-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							
	-BU		BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
E	-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)						E	
	-BZ,-BZ4		MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							
	-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
	-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							
									-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							
									-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
									-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
									-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
									-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							
F					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM WBC OUTGOING		Scale			= H02	F
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS					
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS					
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS					
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS					
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK					
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING					
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH					
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES					
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR					
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES					
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS					
	-PFF	FLAG RELAYS						-RAR	RESISTORS					
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS					
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE					
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION					
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD					
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1					
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2					
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2					
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS					
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4					
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5					
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES					
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS					
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS					
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS					
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS					
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES					
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS					
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS					
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES					
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS					
	-PGV	VOLTMETERS						-TB	CONVERTER					
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE					
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE					
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE					
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE					
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER					
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)												
F	-QBD4	TWO-WAY DISCONNECTOR												
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER												

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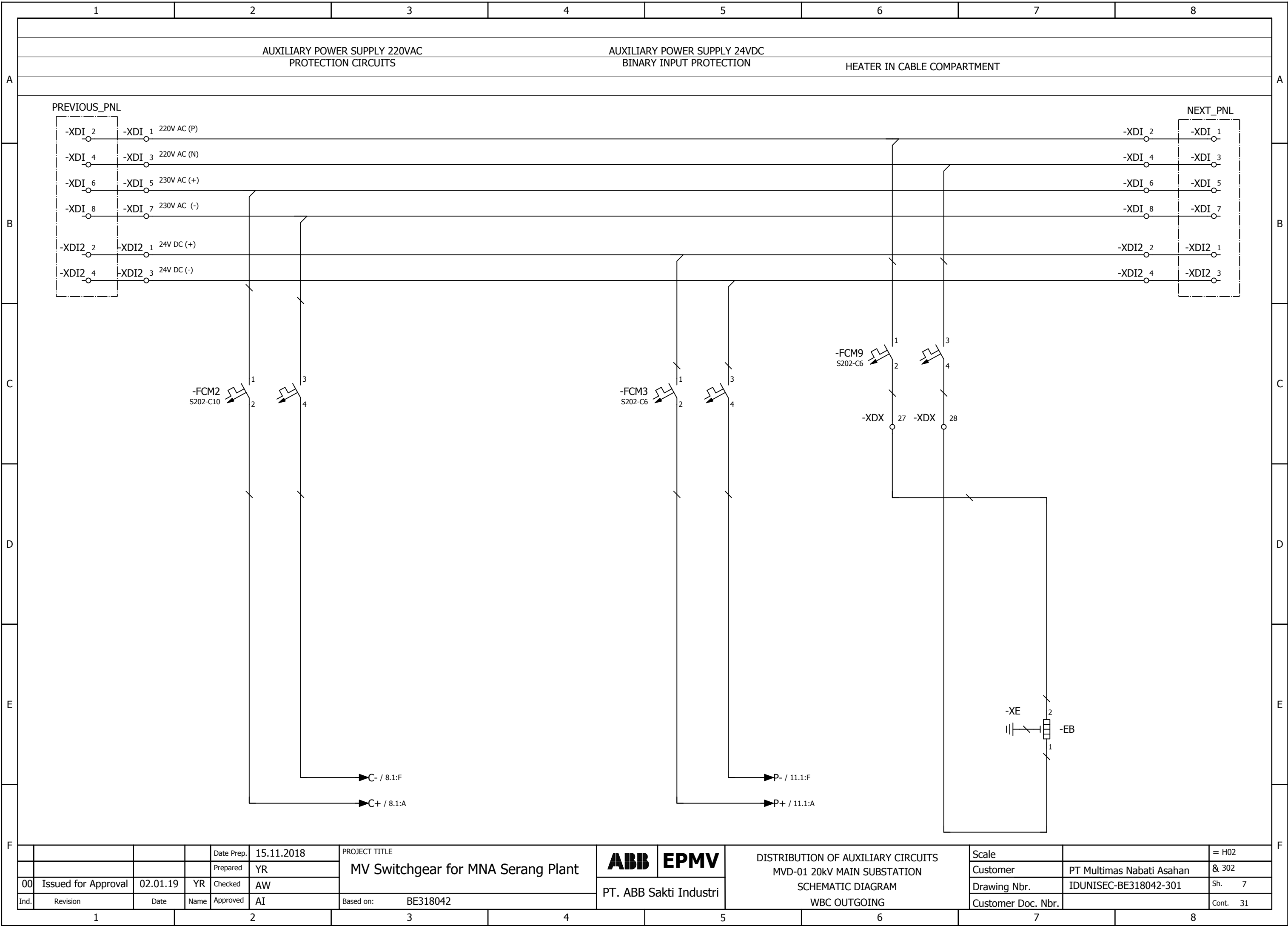
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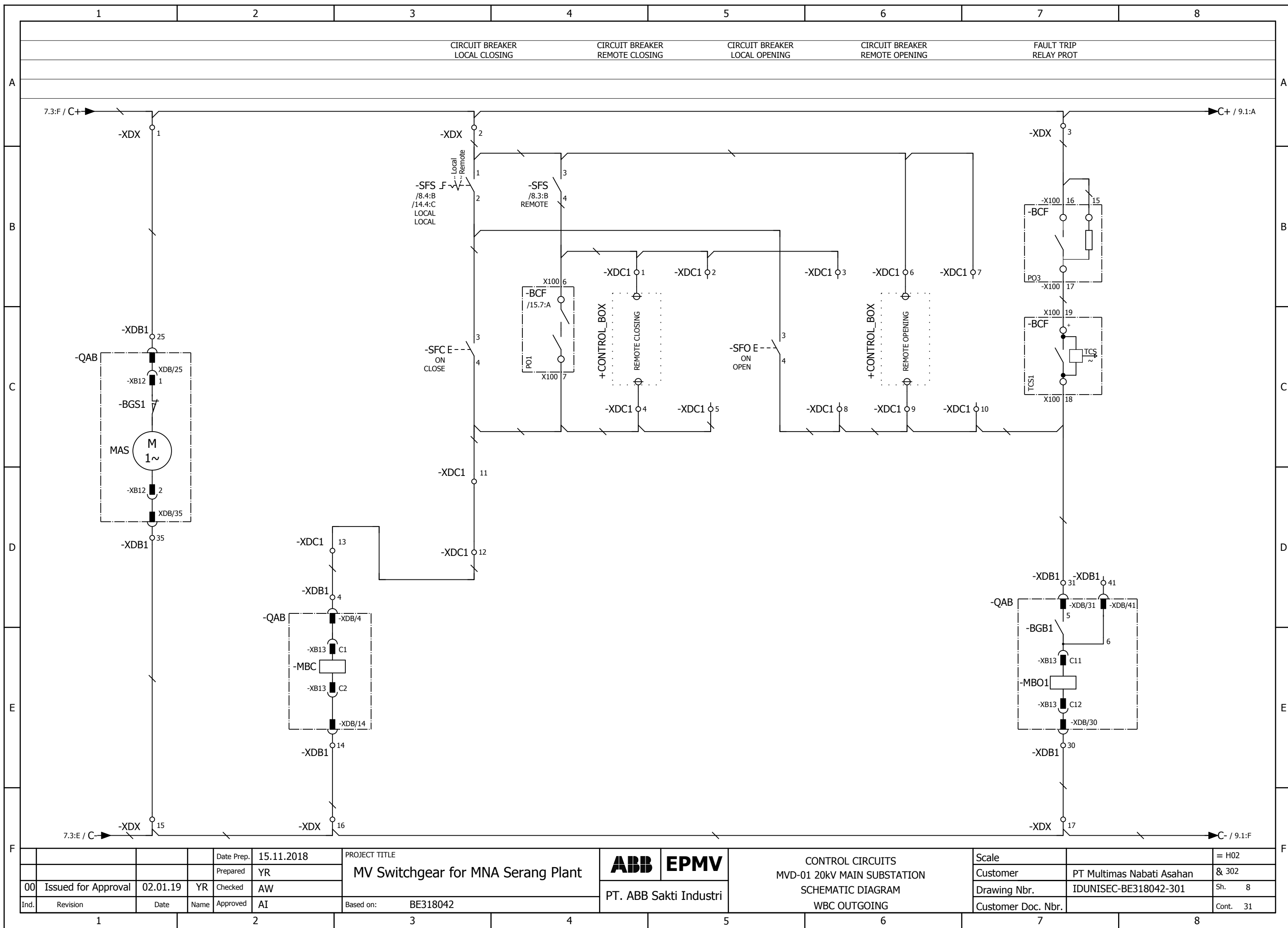
A	-TFA ACTIVE POWER TRANSDUCERS				-XD11 TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS				A
	-TFC CURRENT TRANSDUCERS				-XD12 TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING				
B	-TFF FREQUENCY TRANSDUCERS				-XD13 TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION				B
	-TFJ ACTIVE ENERGY TRANSDUCERS				-XD14 TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS				
C	-TFK REACTIVE ENERGY TRANSDUCERS				-XD16 TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS				C
	-TFM MULTIFUNCTION TRANSDUCERS				-XD17 TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS				
D	-TFP POWER-FACTOR TRANSDUCERS				-XD18 TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS				D
	-TFQ REACTIVE POWER TRANSDUCERS				-XD19 TERMINAL BLOCK FOR SPECIAL USE				
E	-TFS SIGNAL CONVERTERS				-XDJ TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD				E
	-TFV VOLTAGE TRANSDUCERS				-XDJ1 TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS				
F	-WA BUSBARS				-XDM SEALABLE TERMINAL BLOCK FOR MEASUREMENT				F
	-WBC POWER CABLES				-XDS CURRENT SOCKETS				
G	-WE EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.				-XDT TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB				G
	-WGC CONTROL CABLES				-XDU CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK				
H	-WGS COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)				-XDU1 TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK				H
	-WH OPTICAL FIBRES				-XDV TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS				
I	-XDA TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS				-XDV1 CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1				I
	-XDA1 CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1				-XDV2 CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2				
J	-XDA2 CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2				-XDV3 CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3				J
	-XDA3 CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3				-XDV4 CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR				
K	-XDA5 CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER				-XDX SUPPORT TERMINAL BLOCK				K
	-XDA8 CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS				-XE1 EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE				
L	-XDB CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS				-XE2 EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE				L
	-XDB1 CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)				-XE5 EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE				
M	-XDB2 CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER				-XE6 EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE				M
	-XDB10...-XDB89 CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS				-XE7 EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE				
N	-XDB7 CONNECTOR FOR TRUCK POSITION CONTACTS				-XE8 EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE				N
	-XDB91 CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR				-XE9 EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT				
O	-XDB92 CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR								O
	-XDB93 CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR								
P	-XDB95 CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET								P
	-XDC CUSTOMER TERMINAL BLOCK								
Q	-XDC1 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER								Q
	-XDC2 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR								
R	-XDC3 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT								R
	-XDC4 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS								
S	-XDC5 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER								S
	-XDE TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE								
T	-XDE1 TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE								T
	-XDE2 TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2								
U	-XDE3 TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2								U
	-XDF TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS								
V	-XDF1 CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR								V
	-XDF2 CONNECTOR FOR REAR VENTILATOR CIRCUITS								
W	-XDF3 CONNECTOR FOR FRONT VENTILATOR CIRCUITS								W
	-XDH TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)								
X	-XDI TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)								X

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM WBC OUTGOING	Scale		= H02
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 302
00	Issued for Approval	02.01.19	YR	Checked	AW			PT. ABB Sakti Industri		Drawing Nbr.	IDUNISEC-BE318042-301	Sh.	6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				Customer Doc. Nbr.		Cont.	31

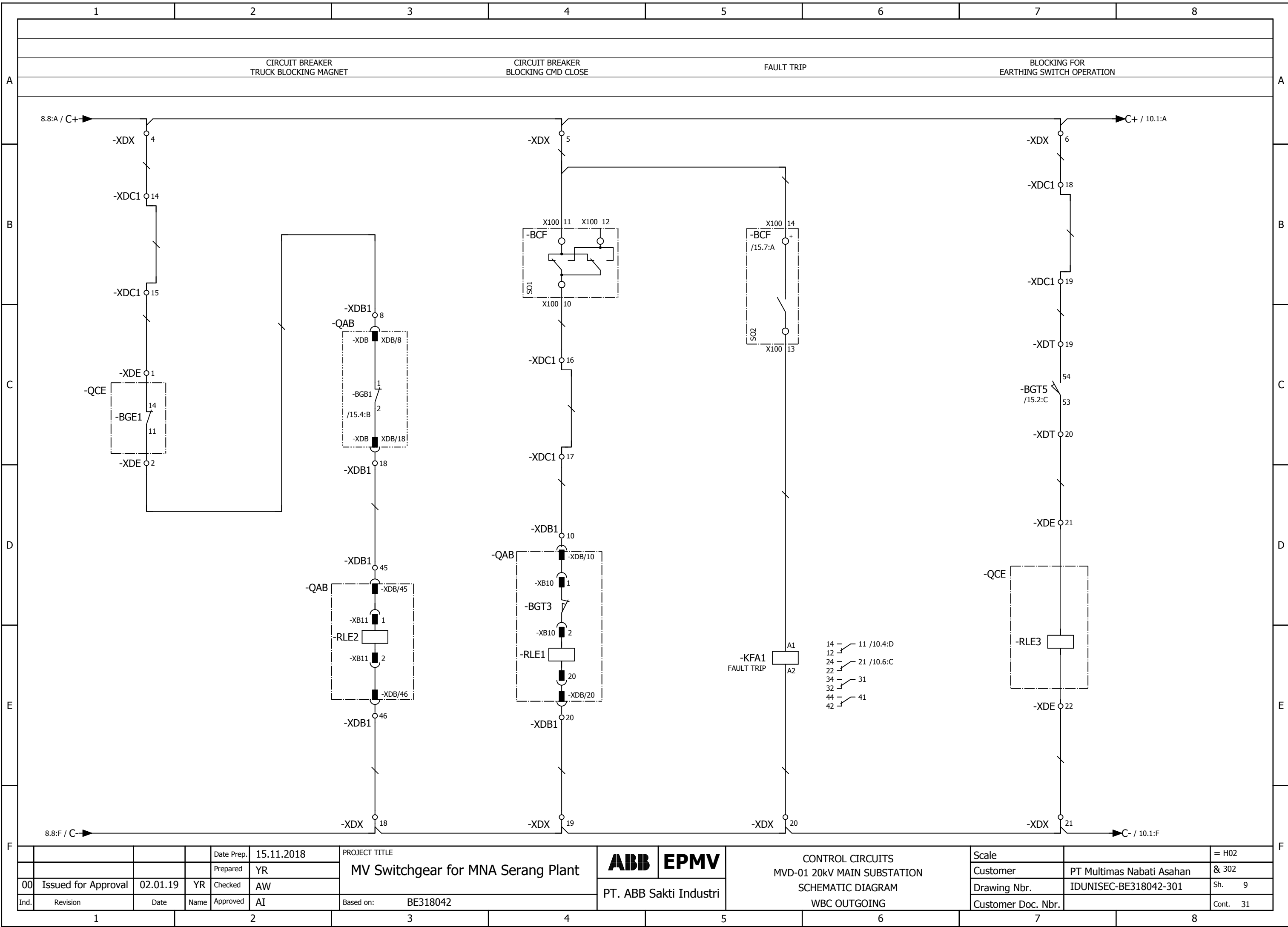
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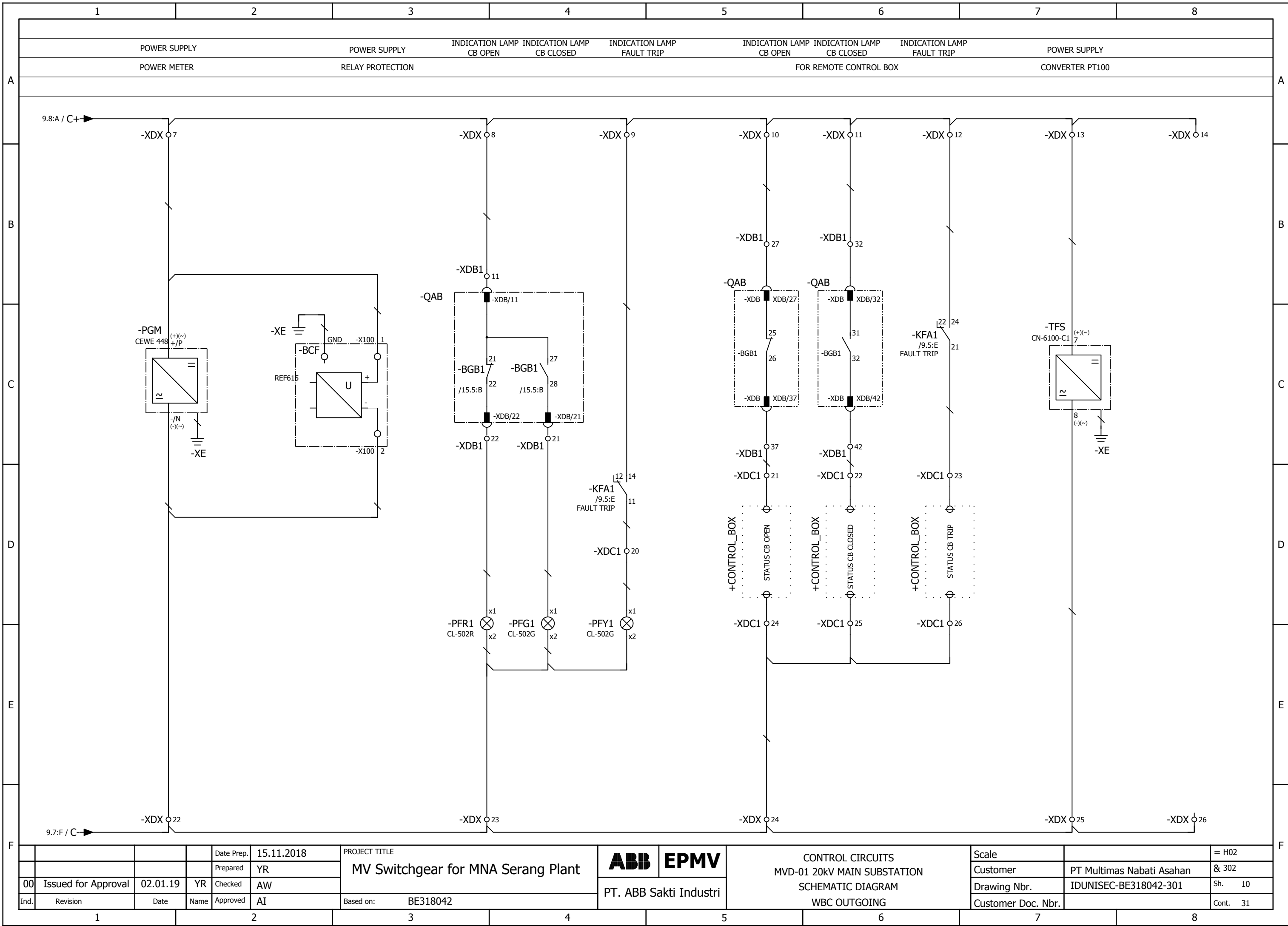




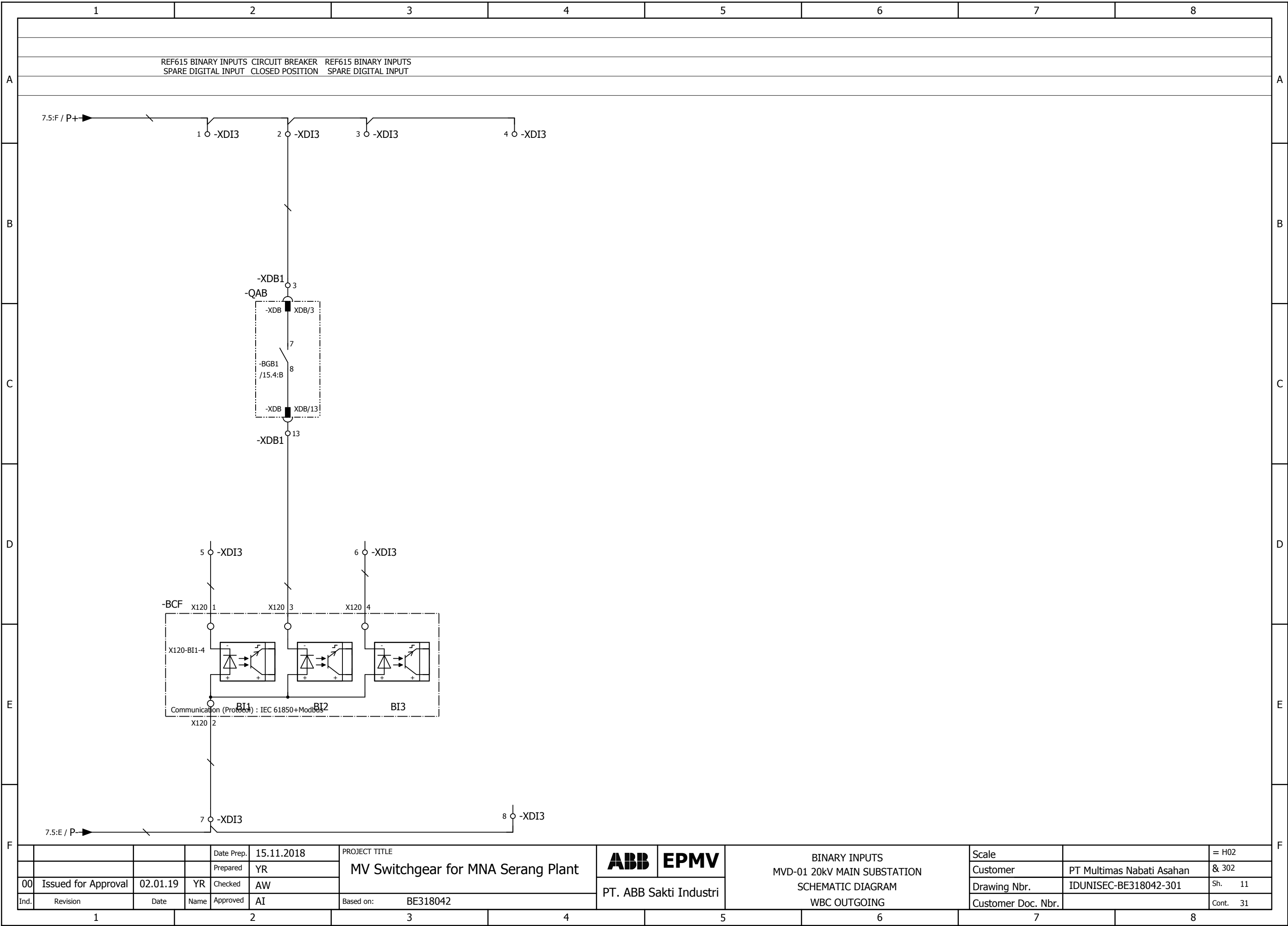
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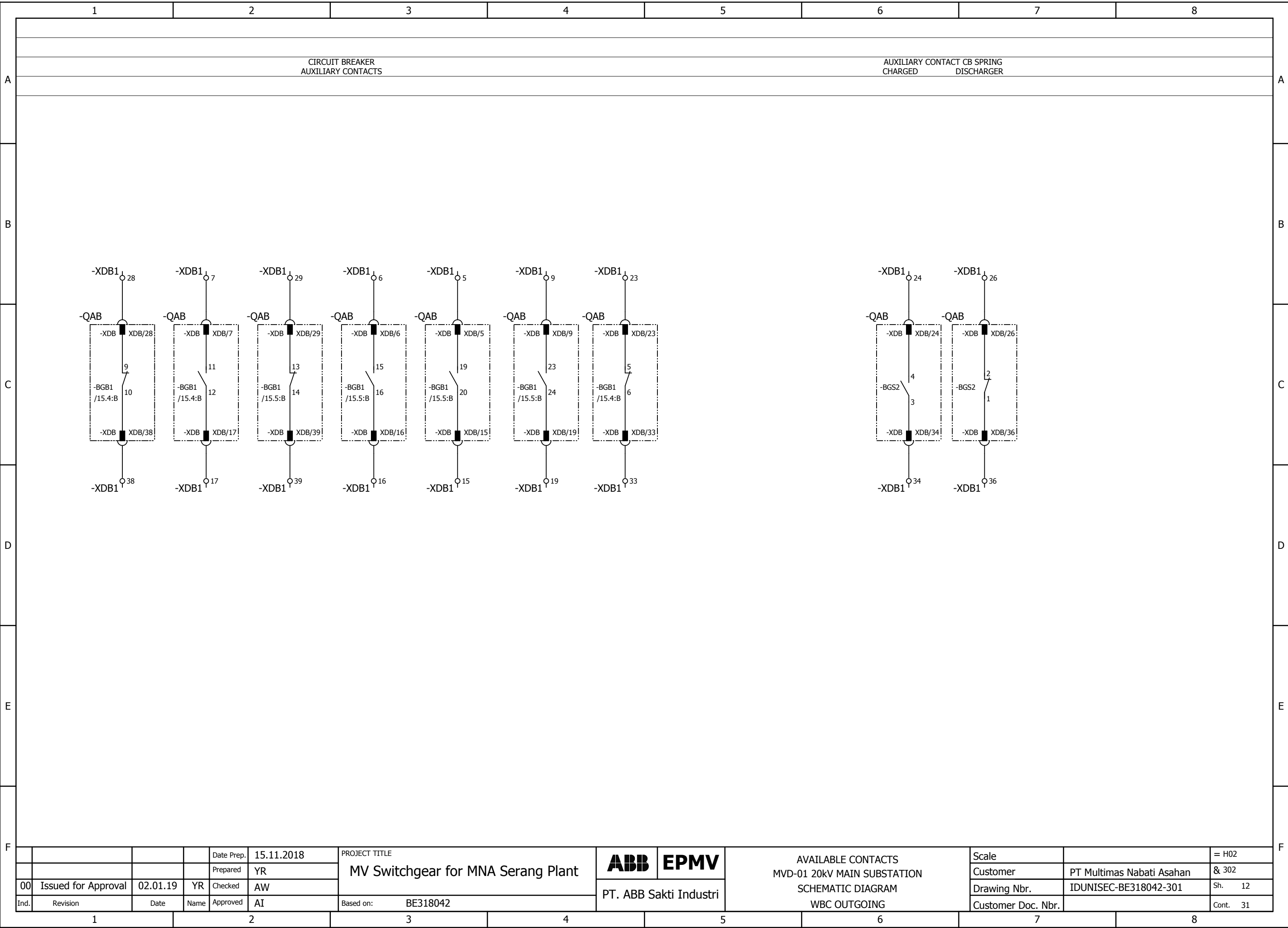


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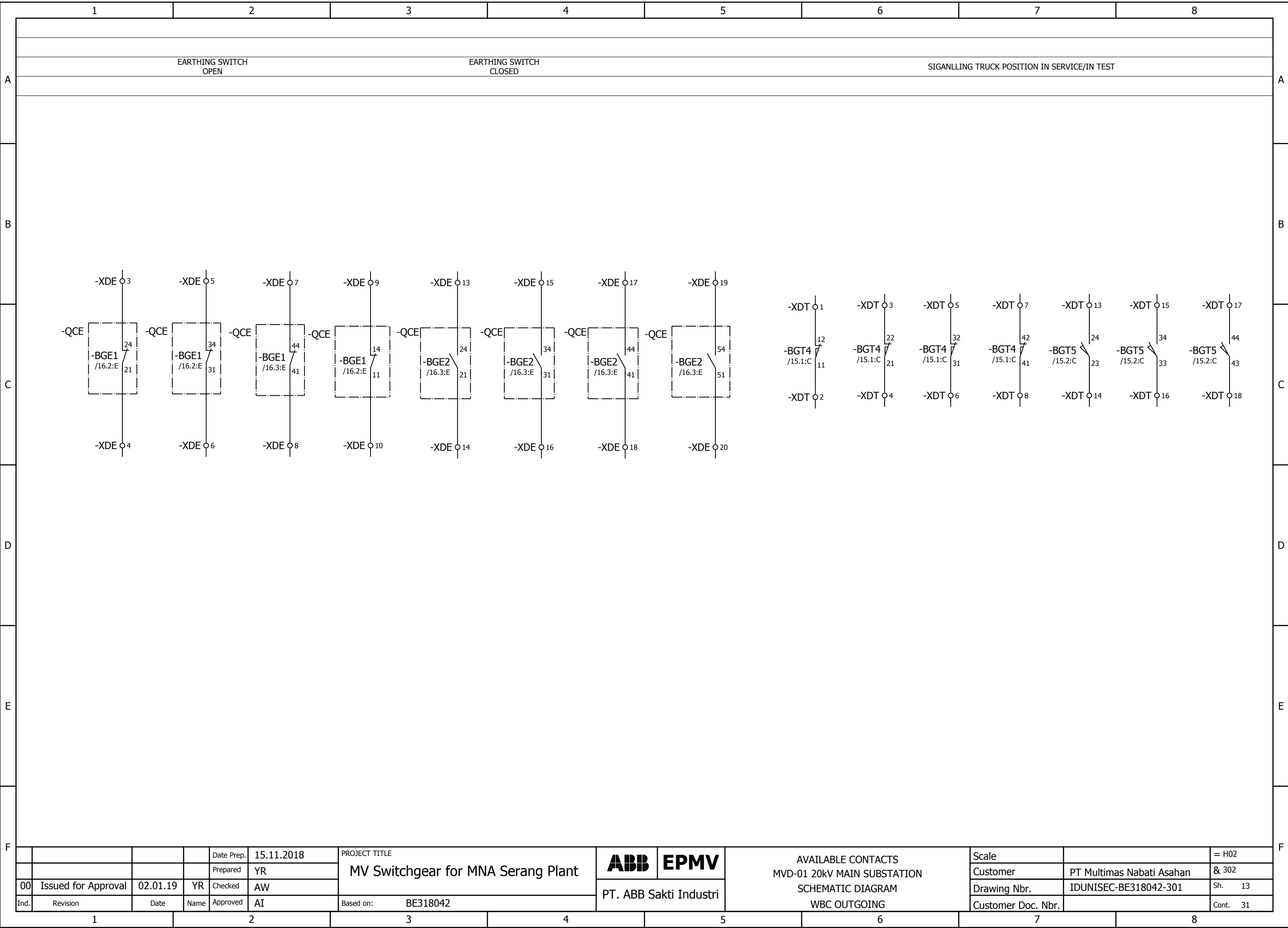


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	BINARY INPUTS MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM WBC OUTGOING	Scale		= H02
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 302
00	Issued for Approval	02.01.19	YR	Checked	AW		Drawing Nbr.	IDUNISEC-BE318042-301		Sh. 11		
Ind.	Revision	Date	Name	Approved	AI	Based on:	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 31
					BE318042							

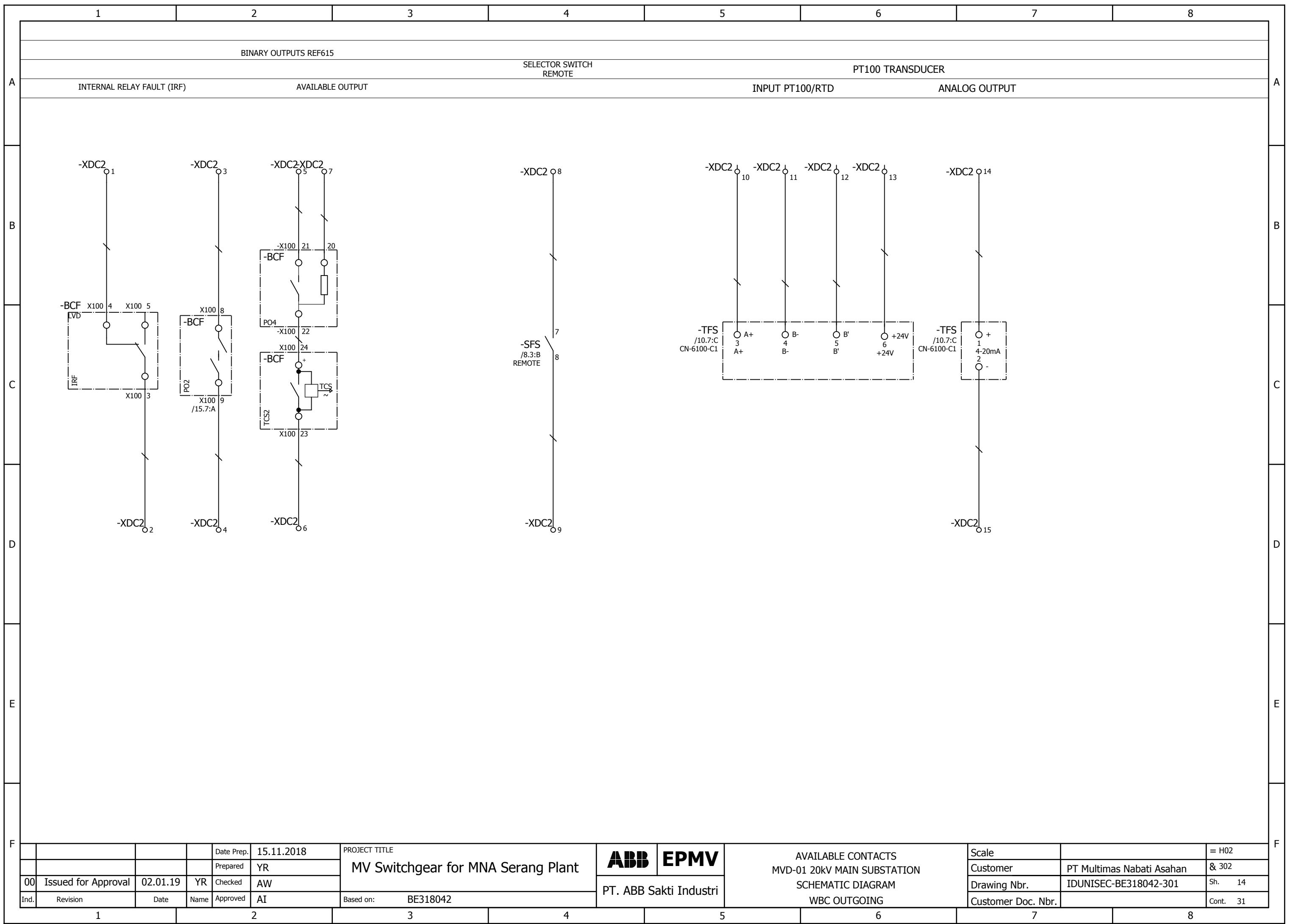
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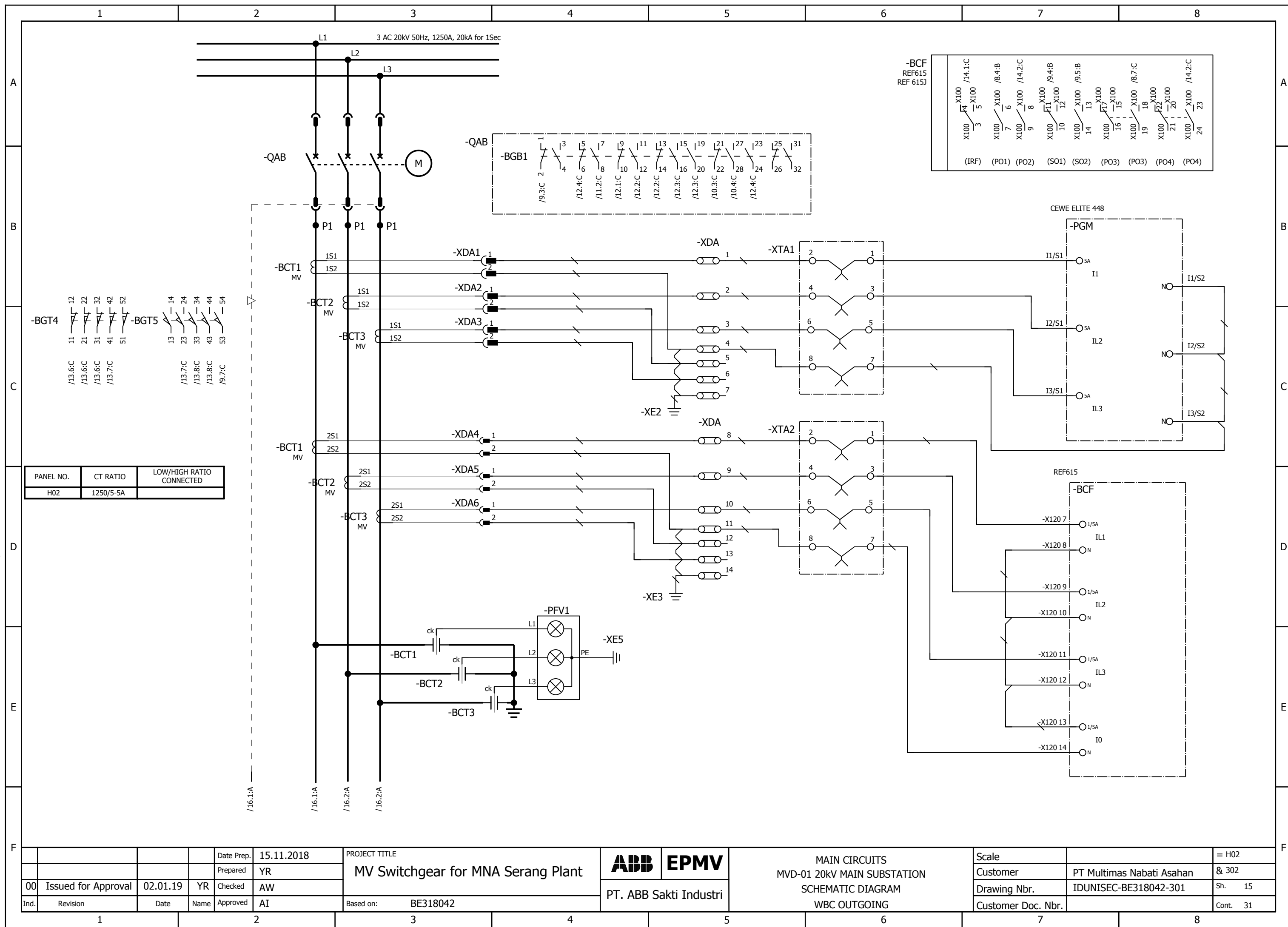
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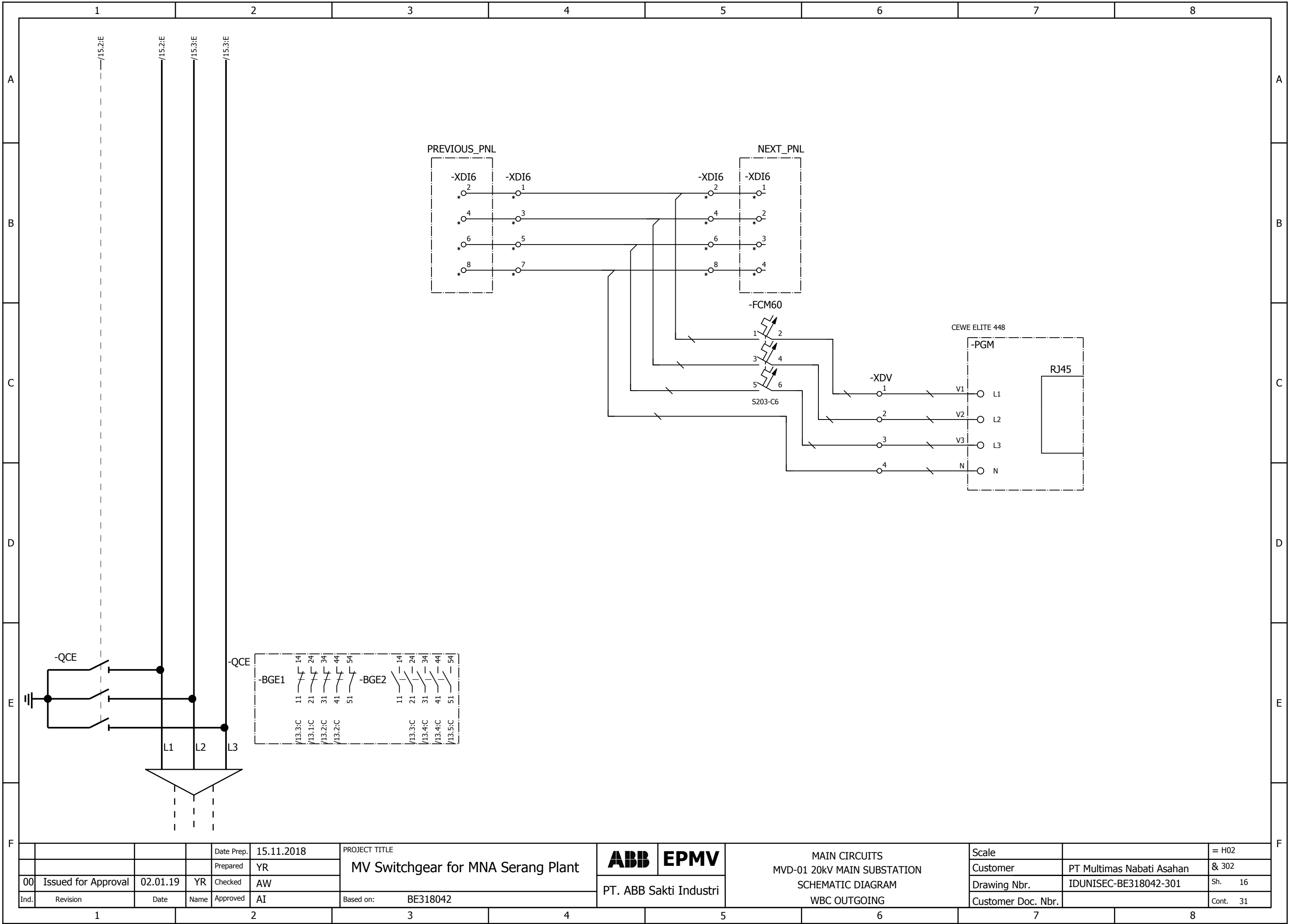
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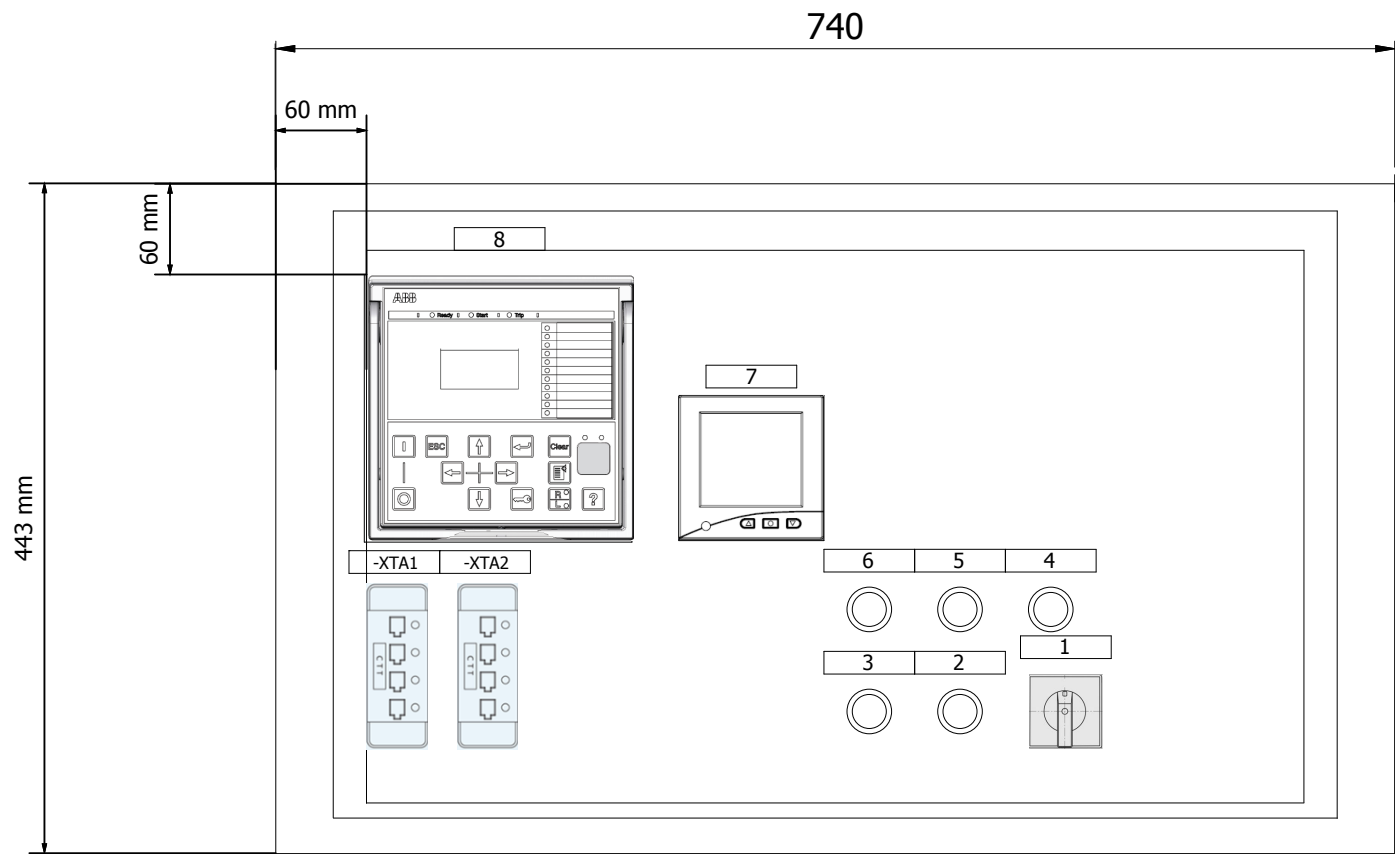


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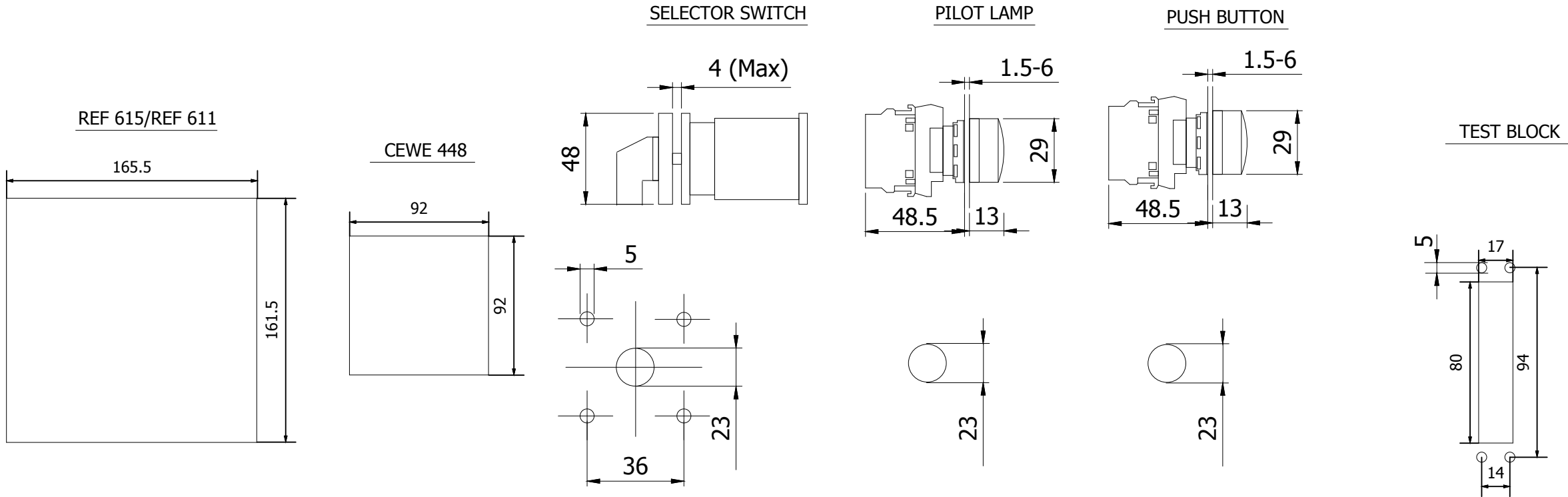
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV	MAIN CIRCUITS MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM WBC OUTGOING		Scale		= H02
				Prepared	YR						Customer	PT Multimas Nabati Asahan	& 302
00	Issued for Approval	02.01.19	YR	Checked	AW						Drawing Nbr.	IDUNISEC-BE318042-301	Sh. 16
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 31

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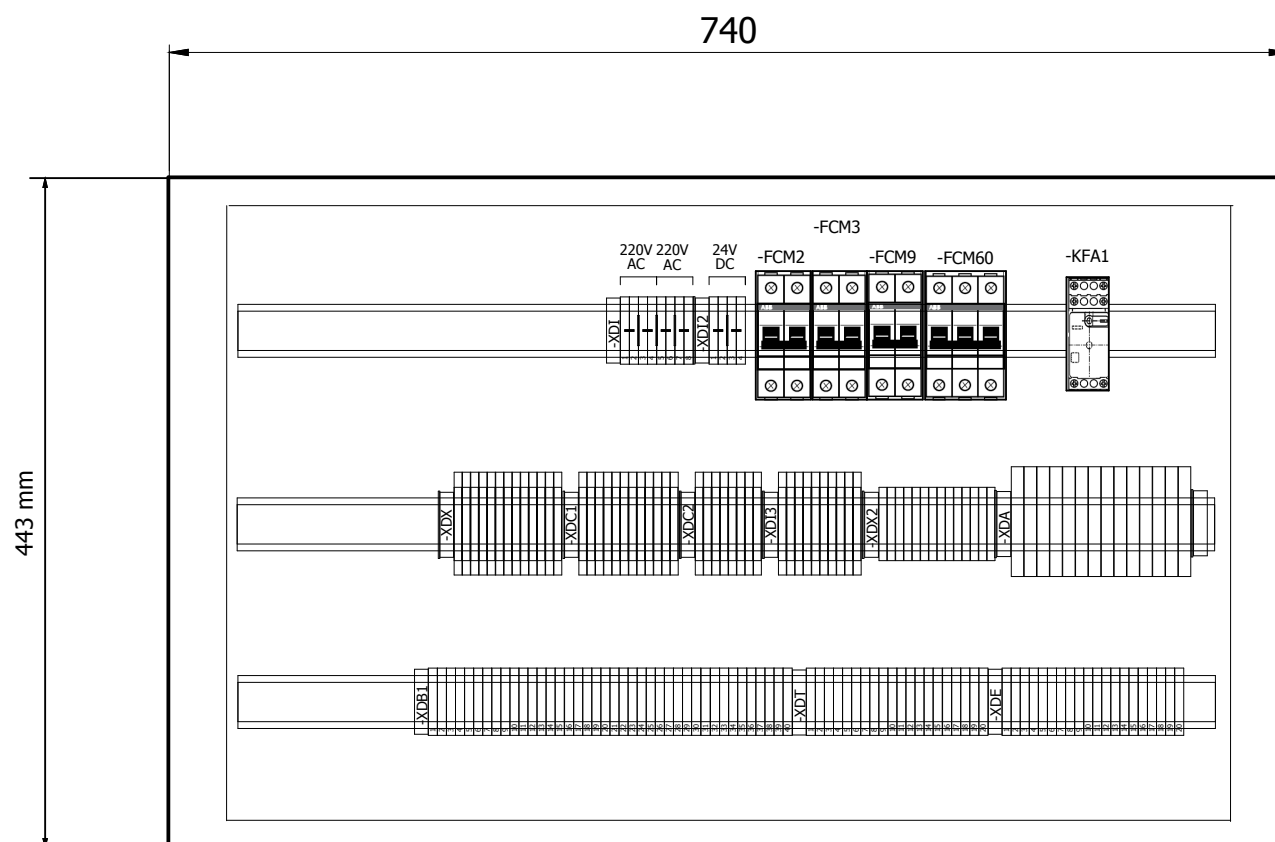
No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

CUT OUT DIMENSION (in mm)



				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV DOOR LAYOUT MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM WBC OUTGOING	Scale		= H02
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 302
00	Issued for Approval	02.01.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		IDUNISEC-BE318042-301	Sh.	17
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.			Cont.	31

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	 	LV BOX LAYOUT MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM WBC OUTGOING	Scale		= H02
				Prepared	YR				Customer	PT Multimas Nabati Asahan	& 302
00	Issued for Approval	02.01.19	YR	Checked	AW				Drawing Nbr.	IDUNISEC-BE318042-301	Sh. 18
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 31

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	1	2	3	4	5	6	7	8						
A	Summarized parts list							IDABB_F02_005-IDABB						
								Carry over: 0.00						
	TAG ID	DESCRIPTION			MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS						
B	-BCF	REF615A HBFAAAAACBC1ANA11G Configuration : A ; Power Supply : 48-250 Vdc; 100-240 Vac			ABB	HBFAAAAACBC1ANA11G	1	/10.2:C						
	-EB	HEATER 230VAC, 50W			SANYANG	CDXN010033P0009	1	/7.7:E						
	-FCM2	MCB S202-C10 MCB. 2P, 10A			ABB	2CDS252001R0104	1	/7.2:C						
	-FCM3;-FCM9	MCB S202-C6 MCB. 2P, 6A			ABB	2CDS252001R0064	2	/7.5:C;/7.6:C						
	-FCM60	MCB S203-C6 MCB. 3P, 6A			ABB	2CDS253001R0064	1	/16.5:C						
C	-KFA1	Auxiliary Relay 230AC 4COS			PHOENIX	1YRA3DL266230001	1	/9.5:E						
	-PFG1	PILOT LIGHT GREEN #CL-523G#230VAC CL-523G			ABB	1SFA619402R5232	1	/10.4:D						
	-PFR1	PILOT LIGHT RED #CL-523R#230VAC CL-523R			ABB	1SFA619402R5231	1	/10.3:D						
	-PFY1	PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y			ABB	1SFA619402R5233	1	/10.4:D						
	-PGM	POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC			CEWE	1YRA4EA0350001	1	/10.1:C						
D	-SFC	PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR			IDEC	1YRA3DA14700001	1	/8.3:C						
	-SFO	PUSH BUTTON, YW1B-M1E10-R RED COLOUR			IDEC	1YRA3DA14700002	1	/8.5:C						
	-SFS	LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)			K&N	1YRA3DA17940011	1	/8.3:B						
	-TFS	Isolated PT100 Converter Model : CN-6100-C1			AUTONICS	CN-6100-C1	1	/10.7:C						
	-XTA1;-XTA2	Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting			YONGSUNG	YSCTT-04C	2	/15.5:B;/15.5:C						
E														
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	PART LIST	Scale		= H02
					Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 302
	00	Issued for Approval	02.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM	Drawing Nbr.		Sh. 19
	Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				WBC OUTGOING	Customer Doc. Nbr.		Cont. 31
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F										E	D	C	B	A